

Experiment 13: Perspective Transformation on Video

AIM

To perform Perspective Transformation on a video.

PROCEDURE

1. Read the video.
2. Read frames one by one.
3. Apply perspective transformation.
4. Display transformed frames.

PROGRAM

```
import cv2
import numpy as np

cap = cv2.VideoCapture(r"C:\Users\chait\OneDrive\Documents\Desktop\video.mp4")

while True:
    ret, frame = cap.read()

    if not ret:
        break

    rows, cols = frame.shape[:2]

    pts1 = np.float32([[0,0],[cols,0],[0,rows],[cols,rows]])
    pts2 = np.float32([[50,50],[cols-50,0],[0,rows],[cols,rows-50]])

    M = cv2.getPerspectiveTransform(pts1, pts2)

    output = cv2.warpPerspective(frame, M, (cols, rows))
```

```
cv2.imshow("Perspective Video", output)
```

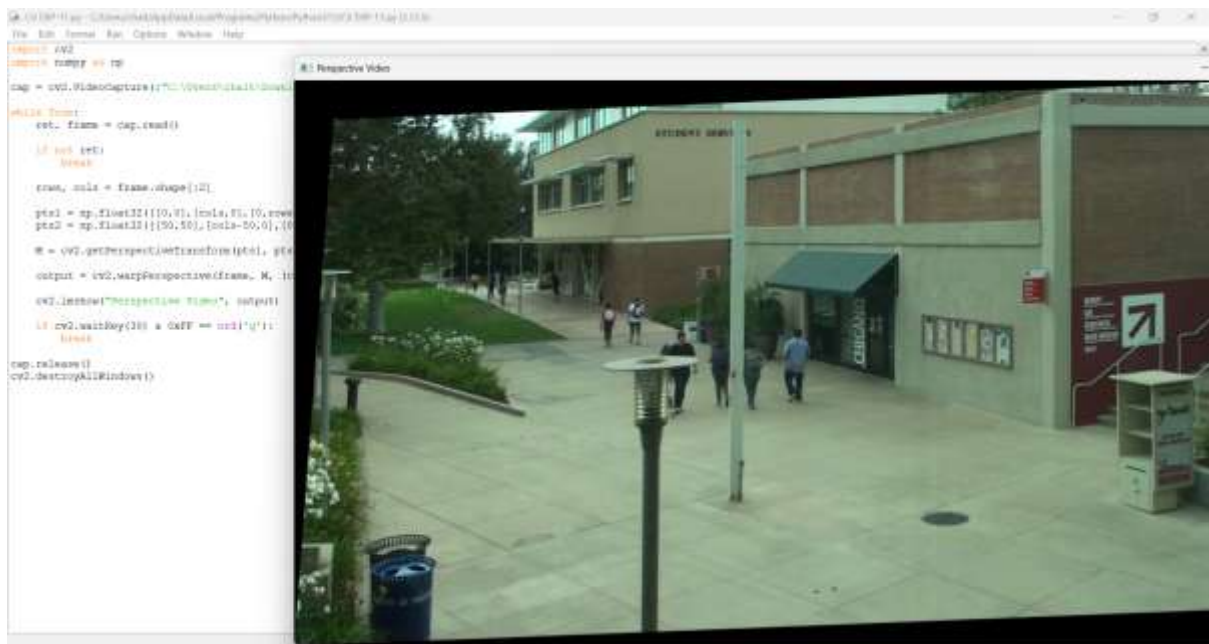
```
if cv2.waitKey(30) & 0xFF == ord('q'):
```

```
    break
```

```
cap.release()
```

```
cv2.destroyAllWindows()
```

OUTPUT:



RESULT:

Perspective transformed video is displayed.